

# Service Skills Training in Restaurants Using Virtual Reality

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## ABSTRACT

Currently, training in restaurants often take the form of on-the-job training (OJT), where trainees are accompanied by trainers who guide them as they serve the customers. However, it is difficult for trainers to understand the cognitive and decision-making processes of the trainees objectively, which makes OJT guidance extremely difficult. Objective measurement indices are also needed. To mitigate these issues, we developed a VR job-training system that focuses on two elements that pose challenges for the actual situation: *awareness* and *priority-setting*.

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## 1 INTRODUCTION

Being an aging society, Japan is currently experiencing a labor shortage, particularly in labor-intensive occupations such as the construction and service sectors. As this is likely to worsen in the future, this study developed support technologies specifically for the hospitality industry (e.g., a restaurant business).

Service training is one area where technology has the potential to save time and help objectively evaluate and improve performance. In restaurant businesses, service training often takes the form of on-the-job training (OJT), where trainees are accompanied by trainers who guide them as they serve customers. However, there are some problems in conducting OJT.

First, a trainer must accompany the trainee, or an expert must spend time to guide the trainee. In addition, when multiple customers are involved, it is difficult to develop trainees' skills, such as *awareness*, which includes service aspects trainees need to pay attention to, and *priority-setting* for various events in multiple locations, which involves trainees being able to prioritize the tasks for an event that requires their immediate attention. This is because it is difficult for the trainers to assess the trainees' cognitive and decision-making processes objectively regarding these skills. Additionally, objective measurement indices are needed to evaluate the effects of training that are not under the influence of a trainer. However, it is difficult to perform "live" assessments in

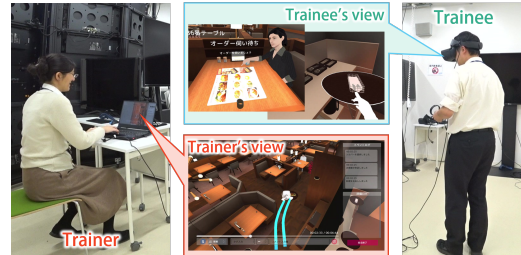


Figure 1: Overview of the proposed system.

an actual situation and consequently, the training efficiency and quality decreases.

To address this, a virtual reality (VR) job training system is herein proposed (Figure 1). In this paper, we have described the design of the proposed system, the implementation of the prototype, and preliminary demonstration.

## 2 RELATED WORK

Many VR training systems have already been proposed to assist trainees in learning procedures, and to acquire the specialized knowledge necessary for their responsibilities at work [Bluemel et al. 2003; Taupiac et al. 2019]. These studies are relevant for learning single-task procedures.

Conversely, a waitron's operations in a restaurant involve performing tasks for multiple tables, simultaneously; awareness and priority-setting are important. Flintham et al. proposed a VR training system for kitchen chefs [Flintham et al. 2020], where the goal was to cook more burgers quickly and to pay attention to the dirt on the floor and wash their hands in a timely manner. Our proposed system is a training system for waitrons aimed at providing service to multiple customers at the right time, while focusing on various aspects. Further, this study also examined how the proposed system provides both qualitative and quantitative feedback to the trainee.

## 3 PROPOSED SYSTEM

The proposed training system must (i) enable training of awareness and priority-setting, and (ii) enable quantitative assessment. However, as it is impossible for a system to quantitatively assess all complex tasks, we added the following feature: (iii) enabling the trainer to qualitatively assess the trainee's performance of a task in real time, or later.

For requirement (i), the system instructs trainees in awareness and priority-setting with regard to events that happen in multiple places simultaneously by simulating a restaurant where multiple customers occupy tables at different times, in the VR space. Different difficulty levels were set so that the trainees could be trained

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progressively. For the prototype implementation, the intervals at which the customers entered the restaurant were used (e.g., a long interval could be set to simulate a low-difficulty slow time). For the training material itself, focus was given to the tasks that are performed in the front-of-house, back-of-house (the employees-only area where the water and cutlery are located), and the line (where completed dishes are picked up), based on actual job duties.

As shown in Figure 1, trainees wore an HMD (Oculus Rift S) and used a controller in each hand for movements and actions in response to events in the VR restaurant. The restaurant 3D model was developed based on the actual restaurant in Tokyo, Japan. Trainees were required to cope with a series of tasks and to observe various areas to perform the tasks at the right time. To ensure that the trainees prioritized the tasks based on the customer's situation, the system shows the progression of meals (Figure 2), and the customers' waiting behaviors when the service was slow (Figure 3).

For requirement (ii), the time required for each task was defined as an index for quantitative assessment of how well the trainee regularly observes the customers, conducts priority-setting, and serves customers without making them wait longer than necessary. In the prototype implementation, the system deducted points when a customer exhibited waiting behaviors and the points were then displayed and shared with the trainee after the training session ended.

For requirement (iii), the system recorded the activity of the trainees, and allowed the trainer to rate their performance and add comments, either in real time, or during a later review (Figure 1). Therefore, trainers would not be required to accompany the trainees. For convenience, trainers were able to play and pause the video freely and a list of the session events, such as customers entering and trainees serving a meal, so that the video could be started at any event by clicking on its name. The proposed system leverages the features of a VR to allow the trainers to evaluate the performance of the trainees from multiple angles by switching between three different perspectives (Figure 4). This feature can also be used by the trainees themselves, to reflect on their own training [Stone 2001].

#### 4 PRELIMINARY DEMONSTRATION

Four participants who were responsible for the waitron training and the curriculum design from the cooperating companies used the prototype system. The experience was received favorably, and the participants commented that the system was suitable for learning the flow of customer service because it made them feel as if they were in an actual restaurant. In particular, the waiting behavior of the customers made the participants experience the same pressure as in an actual restaurant.

However, a "sense of time" was a controversial point. The duration of each event was set to be shorter than that of the actual event to avoid fatigue and VR sickness. However, participants pointed out that it was also important to understand the time required and to anticipate customer's actions; these aspects need to be considered in the future.

#### 5 CONCLUSION

This study proposed a VR job training system to train waitrons in restaurant operations. Focus was given to the training of *awareness* and *priority-setting*, while dealing with events in multiple places



Figure 2: Progression of a meal

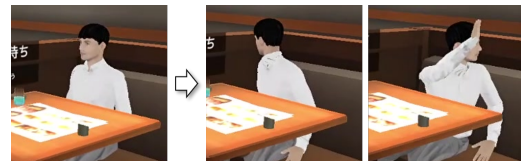


Figure 3: Customer behavior. Left: Normal time, middle and right: when the customer had to wait.



Figure 4: Three different perspectives in scoring mode. From left: Objective, subjective (trainee's point of view), and overhead.

simultaneously, which is otherwise difficult to train in an on-the-job setting in actual restaurants. This system facilitates the feedback of qualitative and quantitative evaluations to trainees.

In future studies, the customer model will need to be updated such that it more closely resembles actual job responsibilities. The method of scoring in awareness also must be improved using an HMD capable of eye tracking, and measuring how often attention needs to be paid to designated places, such as the entrance, tables, etc. User testing also needs to be conducted to evaluate the validity, usability, and usefulness of the proposed system. Finally, the model also must be generalized as a scenario creation tool for use in other restaurants of the same family, or in those restaurants to make the model more versatile.

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